

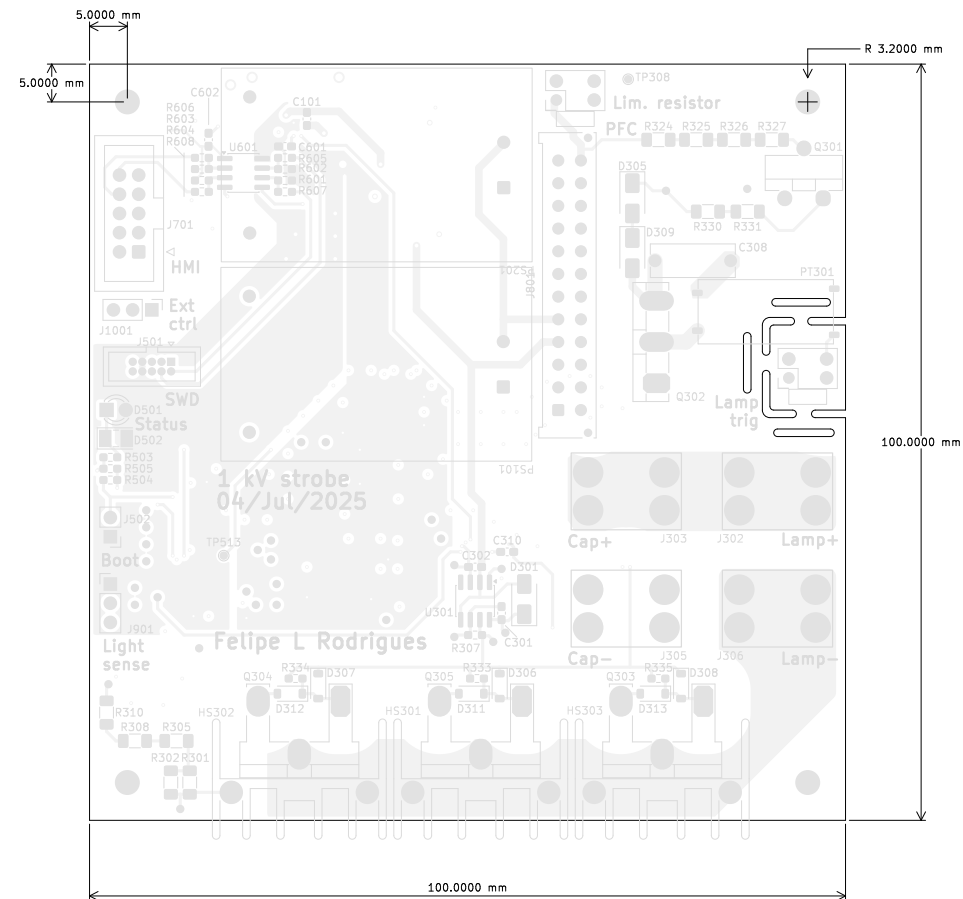
Main board Fabrication Document

Top Fabrication (Scale 1:1)

Layer Stack Legend

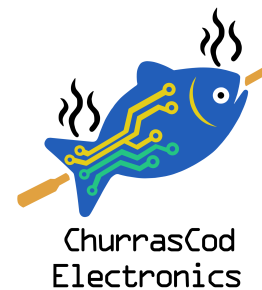
	Material	Layer	Thickness	Dielectric	Type	Gerber
		F.Paste			Paste Mask	
		F.Silkscreen		Direct Printing	Legend	GBR
		F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	Copper	F.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
	Prepreg		0.1mm	FR4_7628	Dielectric	
	Copper	In1.Cu (GND)	0.035mm (1.00oz)		Plane	GBR
	Core		1.828mm	FR4	Dielectric	
	Copper	In2.Cu (Sig, PWR)	0.035mm (1.00oz)		Plane	GBR
	Prepreg		0.1mm	FR4_7628	Dielectric	
	Copper	B.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
		B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
		B.Silkscreen		Direct Printing	Legend	GBR
		B.Paste			Paste Mask	

Total thickness: 2.278mm
Note: external layer thicknesses are specified after plating



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts,GBR" SUFFIX.
- DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.
- SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: HAL SNPB
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:
- A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
- B. Tg 170 C OR EQUIVALENT.
- C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:
- | | |
|------------------|----------------------|
| BOARD SIZE | 100,000 × 100,000 mm |
| BOARD THICKNESS | 2,278 mm |
| TRACE WIDTH | 0,300 mm |
| TRACE TO TRACE | 0,200 mm |
| MIN. HOLE (PTH) | 0,300 mm |
| MIN. HOLE (NPTH) | 1,000 mm |
| ANNULAR RING | 0,100 mm |
| COPPER TO HOLE | 0,300 mm |
| COPPER TO EDGE | 0,500 mm |
| HOLE TO HOLE | 0,250 mm |



Comments:

Company:

ChurrasCod Electronics

Board Name:

Main board

File Name:

Strobe_main.kicad_pcb

Sheet Path:

Variant:

PRELIMINARY

Project Name:

1 kV strobe

Date:

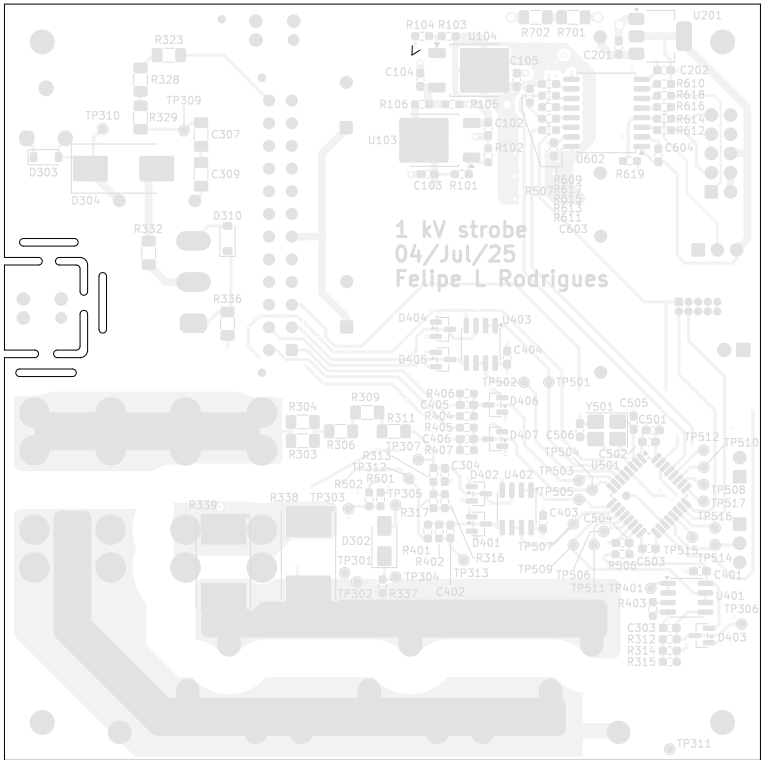
2024-04-13	+ (Unreleased)
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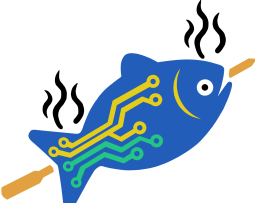
Size:

Sheet: **1** of **10**

Main board Fabrication Document

Bottom Fabrication (Scale 1:1)



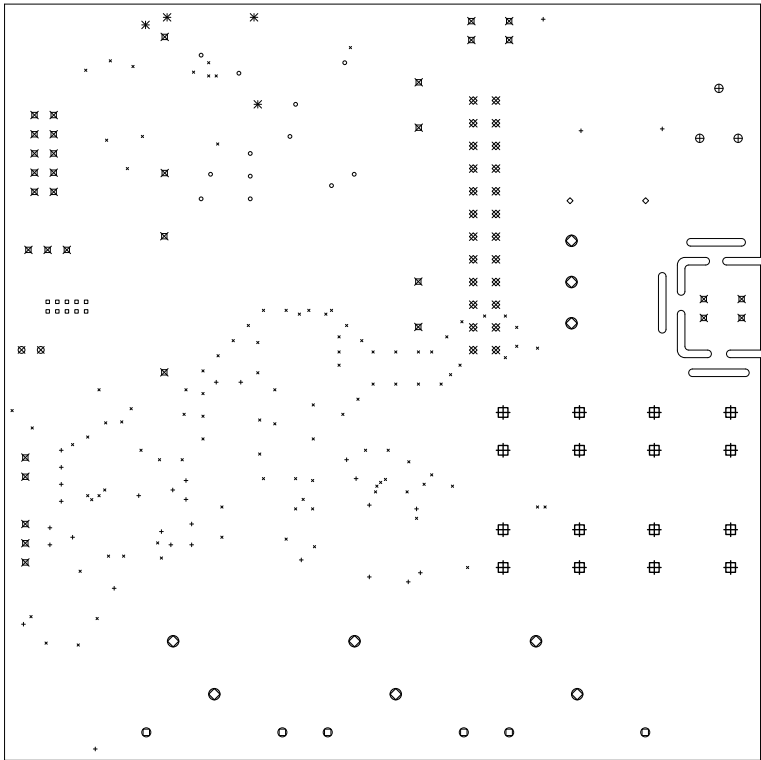
 ChurrasCod Electronics	Comments:		Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: b5afdc7
			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb		Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:			Reviewer:	Size: A3	Sheet: 2 of 10

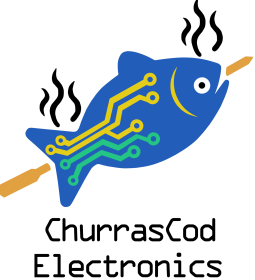
Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	109	0,30mm (11,81mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
○	12	0,50mm (19,69mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
+	31	0,50mm (19,69mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
◻	10	0,65mm (25,59mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
◇	2	0,80mm (31,50mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊗	2	0,90mm (35,43mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
※	4	1,00mm (39,37mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
⊠	34	1,00mm (39,37mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊗	24	1,02mm (40,16mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	3	1,10mm (43,31mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
○	6	1,20mm (47,24mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
○	9	1,50mm (59,06mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	16	1,80mm (70,87mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
	Total 262					



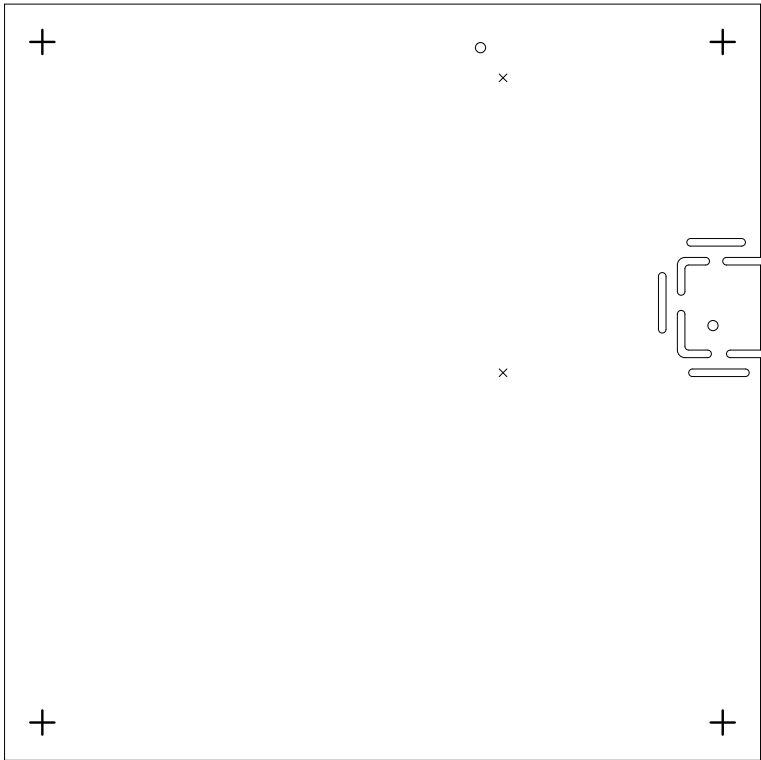
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			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)	
	Sheet Path:		Reviewer:	Size: A3	Sheet: 3 of 10	

Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

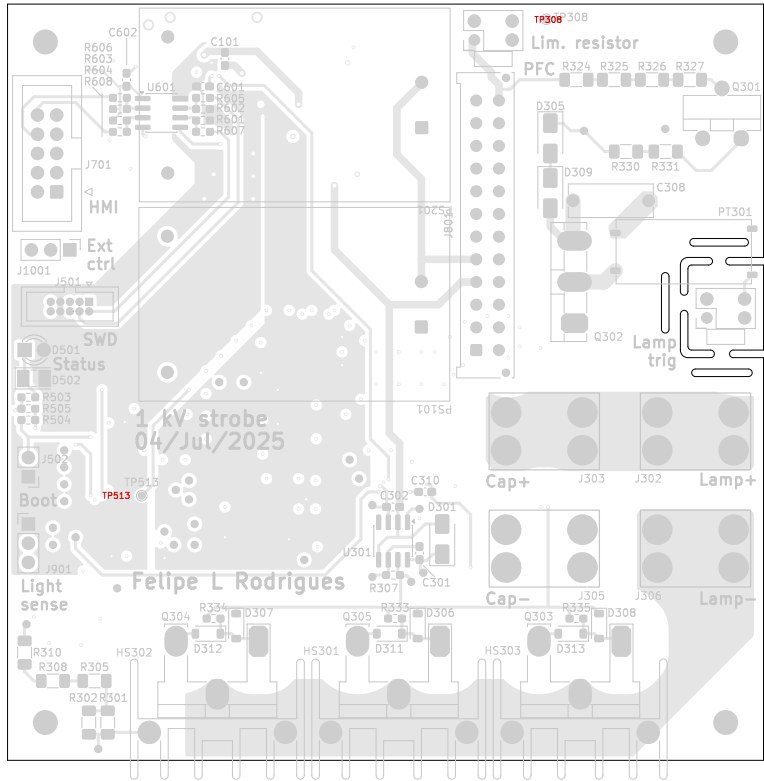
Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	1,02mm (40,16mils)	NPTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Mechanical
○	2	1,35mm (53,15mils)	NPTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Mechanical
+	4	3,20mm (125,98mils)	NPTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Mechanical
Total 8						



	Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: b5afdc7
	Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
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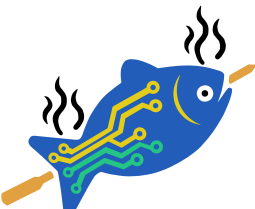
Main board Fabrication Document

Top Test Points (Scale 1:1)



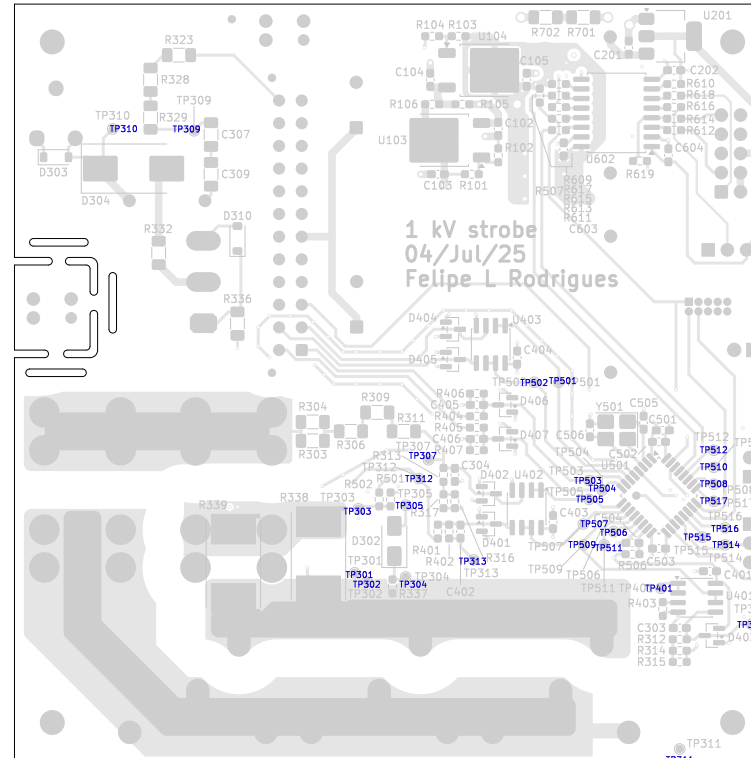
Ref.	Net	X [mm]	Y [mm]
TP308	LAMP_P	76.25	103.00
TP513	Vbus_ctrl	22.75	40.00



 ChurrasCod Electronics	Comments:		Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: b5afdc7
			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Top Test Points (Scale 1:1)	File Name: Strobe_main.kicad_pcb		Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
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Main board Fabrication Document

Bottom Test Points (Scale 1:1)



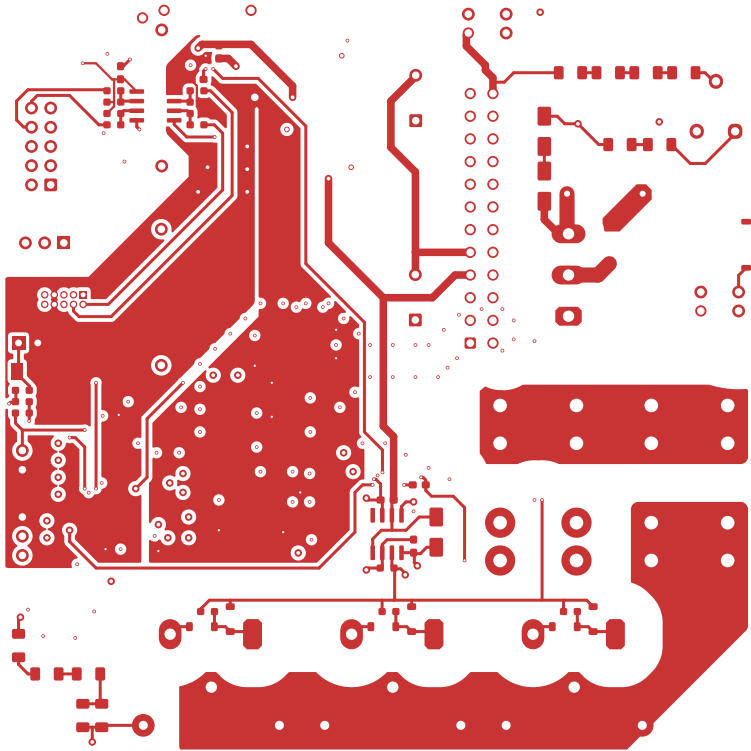
Ref.	Net	X [mm]	Y [mm]
TP301	VBS	60.00	29.80
TP302	DRV_OUT	58.40	28.60
TP303	Lamp_EN	59.50	38.25
TP304	DRV_RET	53.25	29.25
TP305	SH0	53.25	38.75
TP306	Um_Col	7.50	23.00
TP307	Um_Lmp	50.25	44.75
TP309	TP_REF	81.25	88.25
TP310	TP_PRI	92.00	88.50
TP311	LAMP_N	17.00	6.50
TP312	Im_Pulse	51.50	42.25
TP313	SH0	44.25	31.50
TP340	Free_OUT	19.50	27.75
TP501	U_SH1000_buf	33.00	55.00
TP502	U_SH240_buf	36.25	55.00
TP503	I_SH240	29.00	42.00
TP504	I_SH1000	27.25	40.75
TP505	U_L_buf	29.00	39.50
TP506	Net-(U501-PB0)	25.75	35.25
TP507	I_P_buf	29.75	36.25
TP508	Net-(U501-PB3)	12.50	41.50
TP509	U_C_buf	29.75	33.50
TP510	Net-(U501-PB4)	12.50	43.75
TP511	Net-(U501-PA7)	27.00	33.50
TP512	Net-(U501-PB5)	12.50	46.00
TP514	Light_sns	11.00	33.50
TP515	INT_EN	14.00	34.50
TP516	Net-(U501-PA12)	11.00	35.75
TP517	Net-(U501-PA15)	12.50	39.25

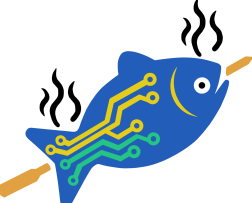


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	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
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Main board Fabrication Document

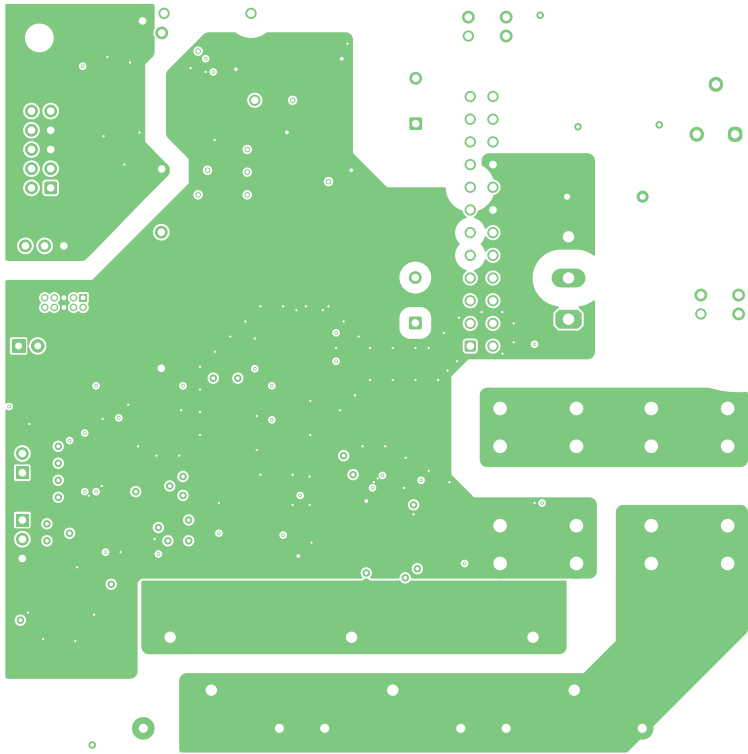
F.Cu (Sig, PWR) (Scale 1:1)

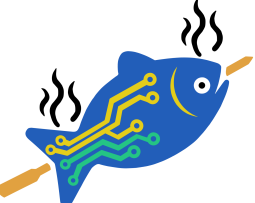


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Main board Fabrication Document

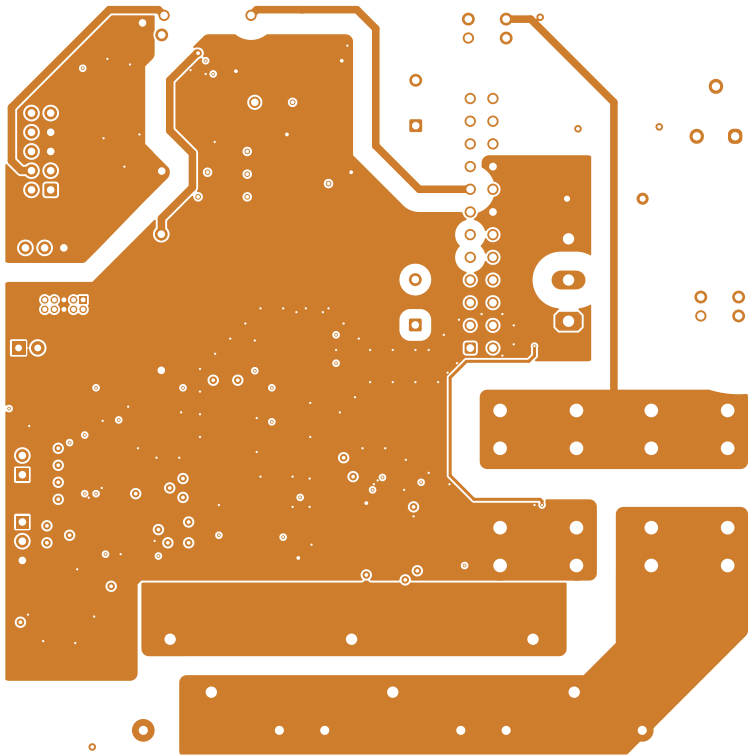
In1.Cu (GND) (Scale 1:1)

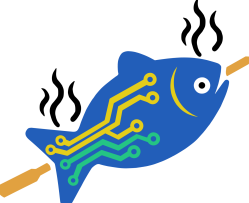


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	Sheet Title: In1.Cu (GND) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
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Main board Fabrication Document

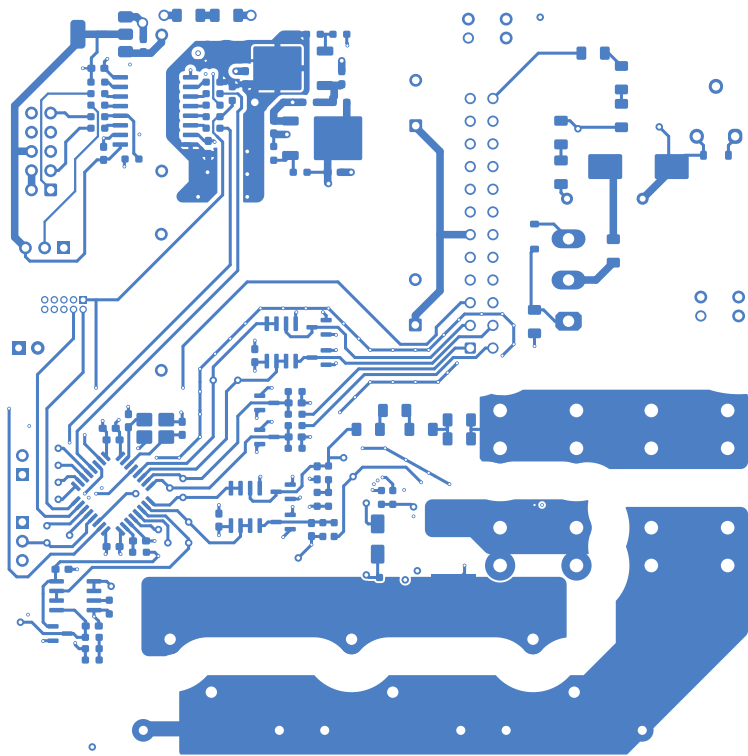
In2.Cu (Sig, PWR) (Scale 1:1)

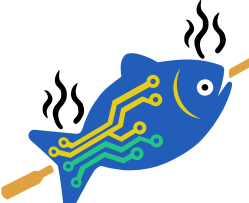


 ChurrasCod Electronics	Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: b5afdc7
	Sheet Title: In2.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 9 of 10

Main board Fabrication Document

B.Cu (Sig, PWR) (Scale 1:1)



 ChurrasCod Electronics	Comments:		Company:		Variant:		Git Hash:		
			ChurrasCod Electronics		PRELIMINARY		b5afdc7		
			Board Name:			Project Name:			
			Main board			1 kV strobe			
Sheet Title:		File Name:		Designer:		Date:		Revision:	
B.Cu (Sig, PWR) (Scale 1:1)		Strobe_main.kicad_pcb		FR		2024-04-13		+ (Unreleased)	
Sheet Path:				Reviewer:		Size:		Sheet:	
						A3		10 of 10	